

From Disembodied Perception to Embodied Agency: Architectural Paradigms, Reasoning Faithfulness, and the Illusion of Generalization in Vision-Language-Action Models

The integration of Vision-Language Models (VLMs) into robotics marks a decisive paradigm shift from reactive, end-to-end policies to deliberative agents capable of explicit reasoning and long-horizon planning. This transition addresses the limitations of static perception-action mappings by decoupling semantic planning from motor control through hierarchical and dual-system architectures [10; RoboOS: A Hierarchical Embodied Framework]. While approaches utilizing Embodied Chain-of-Thought (CoT) enhance interpretability, recent literature reveals a critical tension: explicit textual reasoning often suffers from latency, compounding errors, and a lack of "faithfulness" where actions diverge from generated plans [30]. Furthermore, diagnostic benchmarks expose an "illusion of embodied reasoning," where models rely on lexical-kinematic shortcuts rather than genuine physical understanding [21]. Consequently, the field is converging on hybrid solutions that internalize reasoning into latent spaces for efficiency [Latent Reasoning VLA] and employ physically viable world models to ground cognition in causal dynamics [Physically Viable World Models]. This synthesis argues that achieving robust embodied agency requires balancing the semantic richness of unified foundation models [Pelican-Unify 1.0] with the structural rigor of modular, physically grounded systems to overcome current generalization bottlenecks.

Introduction: The Paradigm Shift from Disembodied Perception to Embodied Agency

The integration of Vision-Language Models (VLMs) into robotics signifies a fundamental transition from static, screen-based pattern recognition to dynamic, goal-directed physical interaction. This evolution marks a decisive departure from early reactive "sense-to-act" policies, which often relied on implicit learning and struggled with generalization outside their training distributions, toward deliberative agents capable of explicit reasoning, long-horizon planning, and self-correction [70; 10]. While initial approaches attempted to map multimodal observations directly to robot actions, creating tightly coupled policies that proved brittle under out-of-domain shifts, recent literature emphasizes the necessity of decoupling semantic planning from motor control while maintaining a unified framework for decision-making [64; 10]. This shift addresses the critical gap between upstream VLM-based reasoning and downstream policy learning, aiming to bridge high-level cognitive capabilities with low-level physical execution [3].

A central theme in this paradigm shift is the move from monolithic end-to-end architectures to hierarchical or dual-system frameworks that explicitly incorporate reasoning steps. Traditional Vision-Language-Action (VLA) models often suffer from "catastrophic forgetting," where fine-tuning on robotic data diminishes the model's foundational world knowledge and reasoning abilities [258]. To mitigate this, researchers have introduced mechanisms such as Embodied Chain-of-Thought (CoT) reasoning, which trains models to generate intermediate representations regarding plans, sub-tasks, and visually grounded features before predicting actions [70; 8]. However, purely semantic reasoning is often insufficient for grounding actions in physical reality; thus, newer frameworks like Embodied Chain-of-Thought (ECoT) and Interleaved Vision-Language Reasoning (IVLR) integrate spatial understanding and visual traces to ensure plans are geometrically feasible and causally coherent [70; 161]. These approaches demonstrate that exposing the reasoning process not only improves interpretability but also enhances performance in complex, long-horizon tasks by allowing for error detection and recovery [174].

Despite these advances, a significant challenge remains in aligning high-level reasoning with low-level execution, a phenomenon sometimes referred to as a lack of "embodied CoT faithfulness," where generated actions fail to match the intended textual plan [30]. To address this misalignment, recent works propose steerable policies and interactive frameworks that allow for human spatial guidance or runtime verification. For instance, the GTA-VLA framework enables users to provide spatial priors, such as affordance points or boxes, to correct grounding errors and steer the policy's reasoning process, effectively transforming the robot from a passive executor into an interactive agent [10]. Similarly, methods like "Do What You Say" employ runtime steering by sampling candidate action sequences and selecting those that best align with the model's own textual plan, thereby turning action diversity into a robustness mechanism [30].

Furthermore, the field is increasingly recognizing the importance of world models that predict semantic state changes rather than raw pixels, facilitating more efficient long-horizon planning [134; 95]. Frameworks such as Agentic Robot and PhysiAgent introduce brain-inspired architectures with standardized action procedures and self-reflection mechanisms, enabling agents to decompose tasks, verify progress, and recover from errors autonomously [174; 125]. These systems often leverage reinforcement learning to refine reasoning-action consistency, moving beyond supervised fine-tuning which can lead to suboptimal policies due to heuristic dataset construction [244; 234].

Nevertheless, contradictions and limitations persist. While some studies argue that explicit CoT reasoning scales reliably with data, others highlight that autoregressive generation of reasoning tokens can introduce latency and compounding errors, suggesting a need for latent planning or CoT-dropout strategies during inference [8; 239]. Additionally, benchmarks like BeTTER and RoboSemanticBench reveal that many state-of-the-art VLAs still rely on lexical-kinematic shortcuts and fail to demonstrate true embodied reasoning when faced with causal interventions or complex semantic grounding tasks [21; 295]. Consequently, the current research trajectory is focused on developing models that not only reason explicitly but also ground that reasoning in physical dynamics and maintain robustness against distribution shifts, ultimately striving for generalist agents capable of operating in unstructured, open-world environments [143; 128].

The Evolution from Reactive Policies to Reasoning Agents

The foundational paradigm of Vision-Language-Action (VLA) models has historically relied on reactive, end-to-end mappings that directly translate perceptual inputs and language instructions into low-level motor commands. While this approach has demonstrated efficacy within training distributions, it frequently falters in novel environments or long-horizon tasks due to an inability to generalize beyond static perception-action correlations [67]. The primary limitation of these reactive systems is their treatment of visual input as a one-shot, static context, which prevents the agent from actively resolving ambiguities or recovering from intermediate execution errors [4]. To overcome these bottlenecks, the field is undergoing a significant shift toward incorporating intermediate reasoning steps, effectively enabling agents to "think before acting" through mechanisms such as Chain-of-Thought (CoT) and recursive planning. This evolution represents a move from disembodied perception to true embodied agency, where reasoning is not merely an auxiliary output but a core component of the control loop.

A critical advancement in this transition is the integration of explicit reasoning traces that decompose high-level goals into executable sub-tasks. Early iterations of this approach utilized text-based CoT, where models generate sequential logical steps before acting. However, research indicates that purely semantic reasoning is often insufficient for robotic policies that require grounding in sensory observations and physical states [70]. Consequently, recent methodologies have evolved to include embodied CoT (ECoT), which grounds reasoning in concrete linguistic action guidance, such as end-effector movements and image-space trajectories, rather than abstract high-level planning alone [8]. Despite these improvements, a persistent challenge remains in the fidelity between the generated textual plan and the executed actions, a phenomenon described as a lack of embodied CoT faithfulness, particularly in out-of-distribution scenarios [30]. To address this, some frameworks now employ runtime verification mechanisms that sample multiple action sequences and select those that best align with the model's own textual reasoning, thereby turning action diversity into a robustness feature.

Beyond text-based reasoning, there is a growing consensus that visual perception must be an active, iterative component of the reasoning process rather than a passive input. The VLA-Thinker framework exemplifies this by introducing a "thinking-with-image" paradigm, where the model dynamically invokes visual tools, such as zooming, during intermediate reasoning steps to gather task-relevant evidence [4]. Similarly, FlowVLA proposes a Visual Chain of Thought that compels the model to reason about motion dynamics—specifically predicting optical flow—before generating future frames, thereby avoiding the "pixel-copying trap" of direct next-frame prediction and ensuring physically plausible forecasts [281]. This interleaving of perception and reasoning allows agents to adapt their visual focus based on evolving cognitive needs, mirroring human cognitive processes more closely than static encoding methods.

To mitigate the high inference latency associated with autoregressive reasoning, researchers are also exploring latent and parallel reasoning architectures. DualCoT-VLA introduces a parallel reasoning mechanism that utilizes learnable query tokens to perform visual and linguistic CoT simultaneously in a single forward pass, significantly reducing latency while maintaining comprehensive multimodal understanding [162]. Complementary to this, Fast-ThinkAct and Latent Reasoning VLA (LaRA-VLA) internalize reasoning into continuous latent representations, distilling explicit CoT capabilities to enable efficient, real-time control without sacrificing long-horizon planning abilities [239; 190]. Furthermore, the integration of world models allows agents to simulate future states and evaluate action consequences internally before execution. The Vision Language World Model (VLWM) leverages language-based abstractions to predict world state changes, enabling a reflective "System-2" planning mode that searches for cost-minimizing action sequences [134].

The evolution also extends to interactive and hierarchical reasoning frameworks that incorporate human guidance and memory. GTA-VLA enables spatially steerable reasoning by allowing users to provide visual priors, such as affordance points, which condition the model's spatial-visual CoT to resolve ambiguities [10]. In navigation tasks, VLingNav employs an adaptive CoT mechanism that dynamically triggers reasoning only when necessary, coupled with a visual-assisted linguistic memory to handle long-horizon spatial dependencies [180]. Additionally, recurrent reasoning frameworks like $\$R^2\VLM process local video snippets iteratively while maintaining a global CoT to estimate task progress, addressing the computational prohibitions of processing long video trajectories [20]. These diverse

approaches collectively signify a paradigm shift where reasoning is no longer an afterthought but a fundamental, dynamic, and multimodal process that bridges the gap between high-level cognition and low-level embodied action.

Defining the Scope of Embodied Cognition and World Models

The contemporary definition of embodied cognition has undergone a profound transformation, expanding from a narrow focus on task completion to a comprehensive framework that integrates spatial understanding, perspective-taking, and the construction of dynamic mental world models. This paradigm shift necessitates that agents move beyond passive perception to actively infer user intent alongside physical states, effectively bridging the gap between disembodied digital reasoning and active physical agency. Recent literature emphasizes that true embodied intelligence requires the synthesis of static scene construction with dynamic interaction, where agents must reason over temporal dynamics and adapt actions in real time [104]. The limitations of prior approaches, which often focused on isolated reasoning skills or static perception, have prompted the development of comprehensive frameworks that evaluate agents across multiple dimensions, including task dynamics reasoning and instant decision-making [104]. Consequently, the field is converging on a definition where the agent acts as a "System 2" cognitive core, capable of high-level reasoning, generalized planning, and failure analysis, rather than merely a reactive policy executor [164].

A critical component of this expanded scope is the ability to construct and utilize "mental world models" that simulate physical dynamics and predict future states. The concept of World Action Models (WAMs) has emerged to address the deficiency in reactive observation-to-action mappings by integrating predictive models of environment dynamics directly into the action generation pipeline [226]. These models target a joint distribution over future states and actions, allowing agents to reason about the consequences of their interventions rather than merely reacting to current observations. Similarly, the WoW project posits that authentic physical intuition must be grounded in extensive, causally rich interactions, demonstrating that generative world models trained on robot interaction trajectories can achieve a probabilistic understanding of physics, albeit with challenges regarding stochastic instabilities [93]. The integration of visual generation into reasoning processes further supports this, as interleaved visual-verbal chain-of-thought reasoning has been shown to outperform purely verbal reasoning in tasks requiring explicit visual modeling of the physical world [211]. This suggests that the scope of embodied cognition now includes the capacity for "thinking with images," where perception is an active, iterative process interleaved with reasoning steps [4].

Spatial understanding within this scope extends beyond basic localization to include complex relational reasoning and coordinate alignment. Benchmarks such as ESPIRE highlight the necessity of evaluating spatial reasoning in physically grounded simulated worlds, decomposing tasks into localization and execution to move from passive reasoning to reasoning-to-act [116]. Furthermore, the ability to align vision-language inputs with spatial coordinates is crucial for managing intricate tasks in complex environments, as demonstrated by approaches that utilize chain-of-thought spatial grounding to harness the reasoning capabilities of language models for advanced spatial tasks [50]. The challenge of spatial hallucination in real-world deployment underscores the need for bio-inspired cognitive frameworks that mimic biological spatial cognition, utilizing dual-map architectures to maintain robust spatial awareness in continuous environments [58]. This evolution is also evident in the move towards point-centric representations, which unify semantic and spatial information to enable zero-shot generalization across diverse robotic embodiments [182].

Perspective-taking and the inference of user intent represent another frontier in defining the scope of embodied cognition. Agents must possess Theory of Mind (ToM) capabilities to infer the mental states of humans, modeling both self and others to generate coherent decisions and actions [1]. This is particularly relevant in human-robot interaction, where models like IntentionVLA are designed to perform implicit human intention reasoning, substantially outperforming baselines in scenarios requiring the interpretation of indirect instructions [43]. The ability to switch between egocentric and world-centric views is also fundamental, as seen in frameworks that enable collaborative spatial reasoning by fusing heterogeneous viewpoints to resolve occlusion and ambiguity [145]. Moreover, the distinction between first-person and third-person perspectives remains a persistent challenge, necessitating datasets and training strategies that explicitly address visual grounding and perspective confusion [82].

Despite these advances, contradictions and gaps remain regarding the efficacy of different reasoning paradigms. While some studies suggest that chain-of-thought prompting offers minimal improvements in spatial tasks that rely on intuitive comprehension [151], others argue that explicit reasoning traces are essential for generalizable robot manipulation, provided they are grounded in concrete linguistic action guidance [8]. Furthermore, the illusion of embodied reasoning in standard benchmarks has been exposed, with evidence suggesting that high success rates may mask fundamental architectural bottlenecks and a lack of true cognitive capability in dynamic scenarios [21].

Consequently, the field is moving towards more rigorous evaluation protocols that decouple high-level reasoning failures from low-level execution limits, ensuring that the scope of embodied cognition truly reflects the complexity of physical agency [21]. The integration of semantic intelligence, inspired by biological cognitive mechanisms, offers a pathway to overcome these limitations by enabling agents to extract and apply deep semantic knowledge of the physical world [223].

Architectural Paradigms: From Unified Foundations to Modular Frameworks

The architectural landscape of embodied artificial intelligence is currently defined by a fundamental divergence between the pursuit of massive, unified foundation models and the development of modular, hierarchical systems tailored for specific physical constraints. On one end of the spectrum, proponents of unified architectures argue that true generalization requires a single model to jointly optimize understanding, reasoning, imagination, and action within a shared semantic space. The Pelican-Unify 1.0 model exemplifies this approach, positing that these capabilities should not be fragmented into independent modules but rather co-evolve through a common training loop where gradients flow across all dimensions simultaneously [121]. Similarly, the HY-Embodied-0.5 family utilizes a Mixture-of-Transformers architecture to enable modality-specific computing while maintaining a unified foundation, demonstrating that efficient edge deployment and complex reasoning can coexist within a single model framework [41]. This philosophy extends to the concept of World Action Models (WAMs), which seek to unify predictive state modeling with action generation, targeting a joint distribution over future states and actions rather than treating them as sequential, disjointed processes [226]. Furthermore, theoretical analyses suggest that despite differences in training data and objectives, language, vision, and action representations may converge toward partially shared semantic structures, supporting the feasibility of such unified models [232].

However, significant challenges persist for purely unified approaches, particularly regarding the conflict between different data domains and the computational demands of real-time control. The OmniActor study reveals that mixing GUI and embodied data in a single model can lead to performance degeneration due to conflicts in deep network layers, necessitating a Layer-heterogeneity Mixture-of-Experts (MoE) design that separates deep parameters while sharing shallow ones [124]. This finding supports the argument for modular or hierarchical designs, such as the "brain-cerebellum" paradigm adopted by RoboOS. In this framework, a multimodal large language model serves as the "Embodied Brain" for global perception and high-level decision-making, while a modular "Cerebellum Skill Library" handles low-level execution, connected by a real-time shared memory mechanism [123]. Such decoupling allows for the integration of diverse embodiments and facilitates multi-agent collaboration without overwhelming a single policy network. Similarly, the BiTAgent framework proposes a bidirectional coupling between MLLMs and world models, establishing complementary pathways for semantic guidance and dynamic feedback to achieve task-aware adaptability [219].

The debate also encompasses the methodological origin of physical priors and the efficiency of reasoning mechanisms. While some systems rely on adapting internet-pretrained models, others argue for training from heterogeneous data sources from the start. The DMO framework, an embodied-native Vision-Language-Action model, integrates web text, autonomous driving scenarios, and embodied interaction logs during pretraining to jointly acquire semantic knowledge and physical priors, avoiding the pitfalls of treating physical grounding as a mere fine-tuning afterthought [27]. In contrast, approaches like ERA demonstrate that smaller models can surpass larger counterparts by distilling foundational knowledge through embodied prior learning and online reinforcement learning, suggesting that the quality and structure of training data may be more critical than model scale alone [26]. Furthermore, the complexity of physical environments often necessitates architectures that can dynamically switch between fast, reactive control and slow, deliberate reasoning. The VLingNav model introduces an adaptive chain-of-thought mechanism grounded in dual-process theory, allowing agents to fluidly toggle between intuitive execution and deliberate planning based on task difficulty [180]. This aligns with the Gemini Robotics 1.5 family, which interleaves internal reasoning processes with action generation, enabling robots to "think before acting" to decompose complex tasks [13].

Theoretical analyses highlight that training cross-morphology agents is a PSPACE-complete problem, suggesting that distributed learning alternatives like Collective Adaptation may be necessary to overcome the limitations of sequential training pipelines [236]. To address the "seeing-to-doing" gap and embodiment heterogeneity, recent works like Embodied-R1 pioneer "pointing" as a unified, embodiment-agnostic intermediate representation, bridging high-level comprehension with generalizable action prediction without requiring task-specific fine-tuning [182]. Meanwhile, other approaches utilize learnable latent codes as bridges to overcome the limitations of natural language interfaces in hierarchical control, enabling flexible communication between high-level planners and low-level policies [152]. Despite these advancements, diagnostic benchmarks like BeTTER reveal that state-of-the-art unified models often fail

under causal interventions, exhibiting behavioral inertia and semantic feature collapse, which underscores the need to balance the scalability of unified foundations with the robustness of modular, physically grounded systems [21]. Future architectural developments may increasingly leverage neurosymbolic approaches to bridge the gap between abstract reasoning and concrete physical laws, ensuring that agents possess both the flexibility of foundation models and the reliability of structured physical understanding [286].

Unified vs. Hierarchical and Dual-System Designs

The architectural landscape of embodied AI is currently defined by a critical tension between monolithic unification and modular specialization, a debate that fundamentally shapes how agents perceive, reason, and act. On one end of the spectrum, unified approaches advocate for a single model capable of handling perception, reasoning, and action simultaneously, positing that injecting action components directly into foundation Vision-Language Models (VLMs) creates natural Vision-Language-Action (VLA) models that leverage strong multi-modal representation learning [67]. Recent work suggests that complexity may not be the primary driver of success in this paradigm; for instance, StarVLA-? demonstrates that a strong VLM backbone combined with minimal architectural design can outperform more complex engineering tricks, achieving state-of-the-art results across multiple benchmarks [139]. Similarly, NORA proposes a smaller, efficient 3B-parameter model that maintains strong task performance while significantly reducing computational overhead, challenging the necessity of massive, monolithic structures for real-time autonomy [84]. Furthermore, methods like VLM2VLA address the issue of catastrophic forgetting in unified models by representing actions as natural language, allowing for parameter-efficient fine-tuning that preserves the model's foundational reasoning capabilities [258].

However, the limitations of purely end-to-end systems, particularly regarding long-horizon planning, error recovery, and cross-embodiment adaptability, have spurred the development of hierarchical and dual-system architectures. These designs typically decouple high-level cognitive reasoning from low-level motor control to mitigate the "illusion of reasoning" often found in static evaluations, where models succeed via lexical-kinematic shortcuts rather than genuine understanding [21]. The RoboOS framework exemplifies this with a "Brain-Cerebellum" architecture, where a multimodal large language model acts as the embodied brain for global perception and decision-making, while a modular skill library handles execution, facilitating robust multi-agent collaboration [123]. This separation allows for specialized optimization; for example, Gemini Robotics 1.5 employs a two-tiered system where a high-level VLM orchestrator plans tasks and delegates execution to a lower-level VLA, enhanced by motion transfer mechanisms to control diverse robot embodiments without robot-specific post-training [13]. Such hierarchical designs are often necessary to bridge the gap between abstract semantic reasoning and precise physical execution, as seen in frameworks like PhysiAgent, which uses VLMs as assistants to organize components based on real-time proficiency feedback from VLAs, thereby overcoming the rigid sequential structures of earlier attempts [125].

A significant subset of recent research focuses on "dual-system" or agentic frameworks that explicitly separate "System 1" (fast, reactive action) from "System 2" (slow, deliberative reasoning). The Agentic Robot framework introduces a Standardized Action Procedure (SAP) that coordinates a large reasoning model, a VLA executor, and a temporal verifier, demonstrating that structured workflows for planning and verification significantly enhance reliability in long-horizon tasks compared to static planning or end-to-end policies [174]. Similarly, OneTwoVLA presents a unified model that adaptively switches between acting and reasoning modes, attempting to capture the benefits of both systems within a single architecture while mitigating the latency and mutual understanding issues often found in strictly separated dual-system approaches [16]. The effectiveness of this separation is further highlighted by RoboBench, which evaluates MLLMs specifically as "embodied brains," revealing that while current models excel in instruction comprehension, they still struggle with spatiotemporal reasoning and failure analysis, suggesting that offloading these complex cognitive tasks to a dedicated reasoning module remains a viable strategy [164].

Despite the theoretical advantages of modularity, integrating these distinct components introduces challenges in communication and grounding. Traditional interfaces relying on natural language instructions can limit the precision with which high-level reasoning steers low-level behavior. To address this, Steerable Policies propose training VLAs on rich synthetic commands at various levels of abstraction, enabling better controllability by off-the-shelf VLMs [7]. Conversely, some researchers argue for tighter integration through intermediate representations. VLA-Thinker, for instance, moves beyond text-based chain-of-thought by treating visual perception as a dynamically invocable reasoning action, creating an interleaved process where the model actively queries visual information during the reasoning phase [4]. This contrasts with approaches like ThinkAct, which compresses reasoning plans into a visual latent space to condition a downstream action model, bridging the gap without requiring explicit natural language handoffs [18].

The debate also extends to the scalability and generalization of these architectures. While unified models like ChatVLA-2 aim to retain open-world reasoning capabilities through mixture-of-expert designs, they risk losing key

competencies during fine-tuning [6]. In contrast, hierarchical systems like EMAC+ dynamically refine high-level textual plans using real-time visual feedback, allowing the reasoning component to internalize environment dynamics rather than relying on static mappings [5]. Furthermore, the emergence of frameworks like RACAS, which uses three communicating LLM/VLM modules to control diverse robots via natural language, suggests that cooperative agentic architectures may offer a more flexible path to robot-agnostic control than retraining monolithic models for every new embodiment [153]. Ultimately, the choice between unified and hierarchical designs often depends on the specific demands of the task, with recent surveys indicating a trend toward hybrid models that leverage the generalization of foundation models while maintaining the structural robustness of modular planning [64] [128].

Hybrid Decoding and Action Generation Mechanisms

The fundamental tension between the sequential, deliberative nature of Chain-of-Thought (CoT) reasoning and the parallelizable, low-latency requirements of high-dimensional robotic action generation has catalyzed a decisive shift toward hybrid architectural paradigms. Traditional autoregressive decoders, while effective for linguistic reasoning, often struggle to maintain the temporal fidelity and reactive speed necessary for precise motor control. This architectural mismatch, where a single decoder attempts to handle both abstract semantic planning and concrete physical execution, frequently results in degraded performance and a weak causal link between thought and action [40]. To resolve this, recent innovations have moved beyond monolithic designs, proposing frameworks that either decouple reasoning and action into specialized modules or integrate them through novel attention mechanisms and latent representations, effectively allowing models to "think" sequentially while "acting" in parallel.

A primary strategy involves modifying the internal decoding mechanism to accommodate distinct processing modes for reasoning and acting within a unified model. DeepThinkVLA addresses the conflict directly by introducing a hybrid-attention decoder that utilizes causal attention for sequential CoT generation before switching to bidirectional attention for the fast, parallel decoding of action vectors. This design significantly improves motor control without sacrificing reasoning depth by respecting the distinct temporal requirements of each modality. Similarly, OneTwoVLA proposes a unified model capable of adaptively switching between "System One" (fast, intuitive action) and "System Two" (deliberate reasoning) modes. By dynamically triggering explicit reasoning only at critical decision points, OneTwoVLA mitigates the latency issues associated with continuous CoT generation while maintaining robust long-horizon planning capabilities. These approaches highlight a growing consensus that rigidly coupling reasoning and action tokens in a single autoregressive stream is suboptimal for complex embodied tasks, necessitating architectures that can fluidly transition between deliberation and reaction.

Parallel to internal architectural modifications, there is a strong trend toward hierarchical and modular frameworks that separate high-level cognitive planning from low-level execution to enhance robustness and generalization. Robix exemplifies this by functioning as a high-level cognitive layer that generates atomic commands for low-level controllers while maintaining natural language dialogue capabilities, enabling the handling of complex instructions and real-time interruptions. Agentic Robot further refines this modularity through a Standardized Action Procedure (SAP), coordinating a large reasoning model for subgoal decomposition, a vision-language-action executor for continuous control, and a temporal verifier for error recovery. Such multi-component systems, including RoboOS with its Brain-Cerebellum hierarchy, demonstrate that decoupling semantic reasoning from physical execution allows for more robust error correction and cross-embodiment generalization than end-to-end policies alone. However, critics of purely modular approaches argue that they can suffer from information loss at the interface between modules, prompting hybrid solutions like EMAC+ that enable bidirectional refinement between high-level textual plans and low-level visual feedback.

To address the inference latency inherent in explicit textual reasoning, several works have explored compressing reasoning traces into latent spaces or visual representations, effectively creating a "latent thinking" paradigm. Fast-ThinkAct and Latent Reasoning VLA (LaRA-VLA) propose distilling explicit CoT into compact latent representations, thereby reducing inference latency by up to 90% while preserving planning efficacy. This approach avoids the token-generation bottleneck of standard CoT by internalizing multimodal reasoning into continuous latent representations that condition action generation. Alternatively, CoT-VLA and Thinking in Text and Images: Interleaved Vision-Language Reasoning Traces for Long-Horizon Robot Manipulation advocate for visual reasoning traces, where the model predicts future image frames or interleaved visual keyframes as intermediate goals. This approach grounds reasoning directly in the visual domain, which is often more aligned with the geometric constraints of manipulation tasks than abstract text. VLA-Thinker extends this by enabling "thinking-with-image," where the model actively queries visual information during the reasoning process, transforming perception from a static input into a dynamic tool for resolving ambiguity.

Despite these architectural advances, challenges remain regarding the faithfulness of generated actions to the reasoned plans. Do What You Say: Steering Vision-Language-Action Models via Runtime Reasoning-Action Alignment Verification highlights that even with correct textual plans, actions may diverge due to embodiment constraints, proposing runtime steering methods to verify alignment between the plan and the executed trajectory. Furthermore, the

efficiency of reasoning is scrutinized in Training Strategies for Efficient Embodied Reasoning, which suggests that while learning to generate reasoning traces improves representation learning, attending to these traces must be optimized to avoid computational overhead. The evolution from unified foundations to these sophisticated hybrid frameworks underscores a maturing understanding that effective embodied intelligence requires a nuanced balance between the deliberative slowness of reasoning and the reactive speed of action, achieved through architectural innovations that respect the distinct temporal and dimensional requirements of each modality.

Mixture-of-Experts and Modular Specialization

To address the competing demands of semantic reasoning, visual foresight, and precise control within embodied agents, recent architectural paradigms have increasingly shifted from monolithic unified models toward modular frameworks employing Mixture-of-Experts (MoE) and specialized routing mechanisms. This transition acknowledges that a single homogeneous network often struggles to simultaneously optimize for distinct cognitive functions without suffering from catastrophic interference or representational bottlenecks. The fundamental rationale for modular specialization extends beyond mere computational efficiency; it directly addresses the interference that arises when heterogeneous tasks are forced into a shared parameter space. [262] identifies interference among heterogeneous tasks as a primary challenge in building generalist agents for open-world environments. To mitigate this, the authors introduce a MoE architecture with task-level routing, allowing the agent to select specific experts for perception, planning, or grounding depending on the current sub-task. This approach contrasts with simple concatenation of expert networks, advocating instead for dynamic activation where distinct capabilities are learned in isolation and routed based on context, thereby preserving the integrity of specialized knowledge domains.

The implementation of modality-specific computing further exemplifies the efficacy of decoupled processing pathways. [41] adopts a Mixture-of-Transformers (MoT) architecture to enable modality-specific computing, incorporating latent tokens to enhance fine-grained visual perception essential for embodied tasks. By separating computational pathways for different modalities, this design yields superior performance on spatial and embodied benchmarks compared to similarly sized unified models, particularly when high-resolution visual details are critical for manipulation. Similarly, [9] instantiates a MoT architecture to decouple semantic reasoning, visual foresight, and action prediction into specialized experts. This design allows for seamless cross-expert collaboration while maintaining distinct processing pathways, enabling the model to perform embodied multimodal chain-of-thought reasoning more effectively than baselines that force all signals into a single latent stream. These findings suggest that while unification is an attractive theoretical goal, functional specialization remains critical for handling the diverse cognitive loads of physical agency.

Beyond static architectural decoupling, recent works explore dynamic routing mechanisms that activate reasoning modules only when necessary, effectively creating a system of "dynamic sparsity." [36] proposes an uncertainty-based adaptive reasoning framework where a lightweight plugin dynamically triggers reasoning based on action entropy. This allows agents to learn difficulty-aware reasoning policies, avoiding unnecessary computation during simple navigation steps while engaging complex reasoning modules only under high uncertainty. This adaptive mechanism bridges the gap between fully modular systems and unified policies, suggesting that the future of architectural design may lie in conditional activation rather than static separation. Furthermore, [16] introduces a model that adaptively switches between "System One" (acting) and "System Two" (reasoning) modes within a single architecture. While technically a unified model, it functionally operates as a dynamic mixture of experts, activating explicit reasoning only at critical moments. This approach challenges the necessity of maintaining separate expert modules at inference time, arguing that a single model can internalize the specialized knowledge of multiple experts through adaptive switching.

The concept of modularity also extends to the integration of external knowledge bases, supporting the argument for modular augmentation over monolithic learning. [291] and [2] both demonstrate that augmenting navigation agents with external knowledge modules significantly improves performance in complex environments. These systems treat knowledge retrieval and integration as distinct modules that interact with the core perception and policy networks, rather than attempting to encode all world knowledge within the model parameters. This modular approach allows for easier updates to the knowledge base and reduces the burden on the primary model to memorize static facts. Additionally, [32] introduces a dynamic question-answer pipeline that tightly couples perception and decision modules through a closed-loop interaction. Here, the language model acts as a critic and probe for the vision model, effectively creating a modular system where perception is not passive but actively guided by reasoning needs. This interleaved design underscores the importance of specialized interaction protocols between modular components to achieve robust embodied intelligence.

Despite the advantages of modularity, some research suggests that specialized reasoning can be distilled into a single model to reduce complexity and latency. [172] proposes a hierarchical knowledge distillation framework that transfers the capabilities of a complex multi-agent system into a versatile multimodal language model. This approach challenges the necessity of maintaining separate expert modules at inference time, arguing that a single model can internalize the

specialized knowledge of multiple experts through distillation. However, the success of such distillation often relies on the initial existence of specialized components, as seen in the hierarchical system used to generate training data. This tension highlights a key debate: whether distinct capabilities should be learned in isolation and routed dynamically or co-evolved within a shared latent space and then compressed. The emerging consensus suggests a hybrid trajectory where training leverages modular specialization to avoid interference, while inference increasingly adopts adaptive, sparse activation patterns to balance performance with efficiency.

Mechanisms of Spatial, Physical, and Chain-of-Thought Reasoning

The evolution of reasoning in embodied agents has fundamentally shifted from treating visual inputs as static context to establishing dynamic, interleaved processes where perception actively shapes reasoning and action. Early Vision-Language-Action (VLA) models often encoded observations once, relying on text-based Chain-of-Thought (CoT) to guide decisions, a paradigm that frequently lacks faithfulness in physical environments because it prevents the model from revisiting the scene to resolve ambiguities or recover from errors [4]. To bridge this perception-reasoning-decision gap, recent frameworks have redefined perception as an explicitly invocable action. For instance, VLA-Thinker introduces a "thinking-with-image" framework where agents dynamically invoke visual tools, such as zooming into specific regions, to gather task-relevant evidence during the reasoning process rather than relying on a single initial encoding [4]. Similarly, PRISM establishes a closed-loop interaction where a language model critiques and probes a vision model with goal-oriented questions, synthesizing sharp, task-driven scene descriptions that overcome the limitations of passive visual outputs [32]. This active engagement extends beyond static images to physical interaction; PhysVLM-AVR demonstrates that in partially observable environments, agents must actively acquire information through sequential physical actions to integrate observations for coherent reasoning, a capability where current models often struggle despite detecting information incompleteness [37].

Integrating spatial and physical reasoning requires grounding high-level semantic understanding in concrete sensory observations, a challenge that standard text-only CoT fails to address due to semantic drift. Embodied Chain-of-Thought (ECoT) addresses this by training models to reason about plans, motions, and visually grounded features like object bounding boxes and end-effector positions prior to action prediction, significantly improving generalization without additional robot training data [70]. However, the form of this grounding is critical; research indicates that effective CoT must ground semantic understanding in concrete linguistic action guidance, such as image-space trajectories, whereas high-level reasoning alone yields marginal gains [8]. To enhance this spatial grounding, SpatialCoT proposes a bi-directional alignment of vision-language inputs with spatial coordinates, harnessing the reasoning capabilities of language models for advanced spatial task planning [50]. Furthermore, neurosymbolic approaches argue for grounding agent reasoning in image schemas-recurring patterns of sensorimotor experience-to bridge the gap between abstract cognition and physical interaction, offering a more human-like conceptual structure for embodied agents [286].

A critical tension exists between the depth of reasoning provided by explicit textual traces and the efficiency required for real-time control. While explicit CoT improves interpretability and allows for human correction, it incurs high inference latency and suffers from compounding errors during autoregressive generation [8]. To mitigate this, researchers have proposed compressing reasoning into continuous latent representations. LaRA-VLA internalizes multimodal CoT into latent space, eliminating explicit token generation at inference time and reducing latency by up to 90% while maintaining performance [190]. Similarly, Fast-ThinkAct employs verbalizable latent planning distilled from a teacher model to achieve compact yet performant reasoning, effectively connecting compact reasoning to action execution [239]. Another strategy involves decoupling deliberate semantic reasoning from low-latency reactive control, as seen in the "Fast-in-Slow" paradigm of Nav-R1, which allows for efficient navigation without sacrificing coherent long-horizon planning [266]. These approaches suggest that the future of embodied reasoning lies not in abandoning explicit thought, but in optimizing its representation to balance fidelity with speed.

Despite these architectural advances, contradictions remain regarding the scalability and faithfulness of embodied reasoning. Some studies indicate that explicit CoT does not scale reliably as an autoregressive action prefix, prompting solutions like CoT-dropout strategies where models leverage reasoning traces during training but predict actions directly at inference to bypass instability [8]. Moreover, standard benchmarks often mask fundamental failures; diagnostic evaluations reveal that state-of-the-art VLAs can exhibit severe lexical-kinematic shortcuts and fail catastrophically in dynamic scenarios despite high success rates on static tasks [21]. Interactive and human-guided reasoning represents a frontier for resolving these spatial ambiguities. GTA-VLA introduces a framework where users provide spatial priors, such as affordance points or boxes, to steer the agent's reasoning process, enabling failure recovery in out-of-domain settings where autonomous grounding fails [10]. This aligns with efforts to integrate human intention reasoning, where models like IntentionVLA are trained to infer implicit human goals through compact embodied reasoning, significantly outperforming baselines in indirect instruction scenarios [43]. Collectively, these

diverse approaches underscore that robust embodied intelligence requires moving beyond passive observation toward active, interleaved, and physically grounded reasoning mechanisms that can adapt to both environmental dynamics and human intent.

Text-Based vs. Multimodal and Latent Reasoning

The evolution of reasoning mechanisms in embodied artificial intelligence has marked a decisive transition from explicit, text-based Chain-of-Thought (CoT) generation to more efficient, multimodal, and latent reasoning paradigms. Early approaches heavily relied on generating textual traces to bridge high-level semantic understanding with low-level robotic control, operating on the premise that "thinking before acting" via natural language could substantially improve generalization. Foundational works demonstrated that equipping Vision-Language-Action (VLA) models with explicit textual CoT allowed for better decomposition of complex tasks and improved robustness in manipulation scenarios [70]. Similarly, methods like NavGPT utilized large language models to perform zero-shot sequential action prediction by explicitly reasoning through textual descriptions of visual observations and navigation history, proving that intermediate reasoning steps are crucial for complex planning [35]. These text-centric frameworks established that decomposing tasks into sub-goals using natural language enhances the agent's ability to handle long-horizon instructions [240]. However, a consensus has emerged regarding the inherent limitations of purely text-based reasoning in dynamic physical environments. The generation of discrete textual tokens introduces significant inference latency, creating a bottleneck for real-time control where low-latency reactive responses are essential [190]. Furthermore, explicit CoT generation can suffer from compounding errors during autoregressive decoding, leading to instability in long-horizon tasks where a single hallucinated token can derail the entire execution plan [8]. Recent analyses have also highlighted the issue of "embodied CoT faithfulness," where generated textual plans may not align with the actual actions executed, necessitating runtime verification mechanisms to ensure consistency [30].

To address the latency and stability issues associated with explicit token generation, recent research has increasingly focused on internalizing reasoning processes into continuous latent spaces. This paradigm shift aims to eliminate the need for verbose textual output while preserving the benefits of structured reasoning. A prominent example is the Latent Reasoning VLA (LaRA-VLA) framework, which unifies multimodal CoT reasoning into continuous latent representations, thereby reducing inference latency by up to 90% compared to explicit CoT-based approaches without sacrificing performance [190]. Similarly, Fast-ThinkAct achieves compact yet performant planning through verbalizable latent reasoning, distilling reasoning capabilities from a teacher model to align manipulation trajectories efficiently [239]. These latent approaches often employ curriculum-based training or distillation strategies to transition models from explicit supervision to latent dynamics, enabling action-oriented control that is both efficient and robust [190]. The efficacy of latent reasoning is further supported by findings that learning to generate reasoning traces improves representation learning, even if the final inference bypasses explicit token generation, suggesting that the act of reasoning refines the policy's internal state rather than just providing an interpretable trace [274]. Additionally, frameworks like ThinkAct utilize reinforced visual latent planning to compress high-level reasoning plans into visual latents that condition downstream action models, demonstrating that latent spaces can effectively bridge high-level planning with low-level execution [18].

While latent reasoning addresses efficiency, another critical limitation of text-only approaches is their inadequacy in representing complex spatial and physical relationships. Natural language often fails to capture the nuances of 3D geometry and dynamic state changes required for precise embodied interaction. Consequently, there is a growing trend toward multimodal reasoning that integrates visual, symbolic, and geometric modalities directly into the reasoning process. The Chain-of-Symbol (COS) prompting method, for instance, represents spatial relationships with condensed symbols rather than natural language, significantly outperforming standard CoT prompting in spatial planning tasks by reducing token usage and improving accuracy [293]. Extending this concept, SpatialCoT advances spatial reasoning by aligning vision-language inputs with spatial coordinates and harnessing language models for spatial grounding, demonstrating superior performance in navigation and manipulation [50]. Moreover, the "thinking-with-image" paradigm proposed in VLA-Thinker treats visual perception as an explicit, dynamically invocable reasoning action, allowing models to actively query visual information during the reasoning process rather than relying on static visual context [4]. This approach mirrors human cognitive processes where perception and reasoning are interleaved, enabling agents to resolve ambiguities and recover from errors more effectively than text-only systems.

The divergence between text-based and latent/multimodal reasoning also highlights a tension between interpretability and efficiency. Explicit textual CoT offers high interpretability, allowing humans to trace the agent's decision-making logic and correct failures through natural language or spatial guidance [10]. In contrast, latent reasoning sacrifices some

degree of transparency for speed, though recent works attempt to bridge this gap by ensuring latent states remain "verbalizable" or by using hybrid architectures that decouple slow reasoning from fast action execution [239]. Furthermore, the integration of reinforcement learning has proven vital in refining both textual and latent reasoning strategies. Frameworks like Nav-R1 and Embodied-R1 utilize reinforcement learning to optimize reasoning traces and pointing capabilities, respectively, ensuring that reasoning outputs are not only logically sound but also grounded in successful physical execution [266] [182]. The emergence of unified models like OneTwoVLA suggests a future where agents adaptively switch between explicit reasoning for complex planning and latent or direct action for routine tasks, effectively balancing the trade-offs between cognitive depth and operational speed [16]. Ultimately, the field is moving towards hybrid systems that leverage the semantic richness of language, the precision of symbolic and geometric representations, and the efficiency of latent spaces to achieve robust embodied intelligence.

Implicit vs. Explicit Reasoning Traces and Faithfulness

The architectural design of embodied agents hinges on a critical trade-off: whether reasoning traces should be explicitly generated as intermediate tokens during inference or implicitly learned within the model's continuous latent space. Proponents of explicit reasoning, particularly through Chain-of-Thought (CoT) prompting, argue that decomposing instructions into sub-goals enhances interpretability and generalization. For instance, purely language-model-based agents can perform high-level planning by explicitly tracking progress and decomposing tasks, as demonstrated in navigation scenarios where agents identify landmarks and adapt plans [35]. Similarly, training Vision-Language-Action (VLA) models to reason about plans and visually grounded features before predicting actions has been shown to significantly increase success rates in generalization tasks [70]. This approach is further refined by frameworks that treat perception as a dynamically invocable action, allowing models to actively query visual information during the reasoning process rather than relying on static context [4].

However, the reliance on explicit textual traces introduces significant challenges regarding faithfulness, efficiency, and error propagation. A critical issue is that explicit CoT does not scale reliably as an autoregressive action prefix; early reasoning mistakes often compound, leading to failed executions even if the initial plan appears sound [8]. Furthermore, a phenomenon termed "embodied CoT faithfulness" describes a disconnect where correct textual plans fail to align with generated actions, particularly in out-of-distribution scenarios, suggesting that explicit traces can sometimes serve as post-hoc rationalizations rather than genuine drivers of behavior [30]. This misalignment is exacerbated by the fact that standard benchmarks often mask these failures, allowing models to optimize for sensorimotor priors rather than true causal reasoning [21]. To address these latency and faithfulness issues, recent works advocate for implicit reasoning mechanisms that distill reasoning into compact latent representations. Approaches such as verbalizable latent planning achieve substantial reductions in inference latency while maintaining performance by internalizing multimodal CoT into continuous spaces, eliminating the need for explicit token generation at inference time [239]. Similarly, internalizing reasoning into latent spaces better matches the continuous nature of perception and control, avoiding the discretization bottlenecks of text [190].

The modality of the reasoning trace itself is as crucial as its visibility, with evidence suggesting that natural language is not always the optimal representation for spatial and physical dynamics. In domains requiring rigorous spatial manipulation, representing relationships with condensed symbols during intermediate steps often surpasses natural language CoT, reducing token usage while improving accuracy [293]. For physical dynamics, the "visual superiority hypothesis" posits that visual generation serves as a more natural world model than verbal reasoning, which faces representational bottlenecks when describing complex physical states [211]. This is supported by paradigms that explicitly reason about optical flow as an intermediate step before generating future frames, thereby bridging the gap between passive observation and active policy learning more effectively than direct next-frame prediction [281]. Such visual or symbolic intermediates help mitigate spatio-temporal hallucinations, where models otherwise rely on superficial shortcuts rather than genuine causal understanding [146].

Hybrid and adaptive approaches are increasingly emerging to leverage the strengths of both paradigms, dynamically switching between fast, implicit intuition and slow, explicit deliberation based on task uncertainty. Models employing adaptive CoT mechanisms trigger explicit reasoning only when confidence is low, balancing computational efficiency with the need for verifiable decision-making [180]. Additionally, unified paradigms argue that understanding, reasoning, imagination, and action should be interdependent dimensions within a single adaptive loop, where reasoning is mutually constrained by action and imagination to ensure faithfulness to executable outcomes [121]. Despite these advances, challenges remain in grounding these reasoning traces in robust physical intuition; supervised fine-tuning alone may not yield generalizable physical rules without active environmental interaction, highlighting the need for training strategies that enforce genuine causal understanding over static pattern recognition [69]. Consequently, the field is moving toward architectures that can self-correct and refine physical knowledge through iterative loops, stabilizing dynamic perception and consolidating verified insights for future reuse [111].

Spatial Reasoning and Physical Grounding

Spatial reasoning serves as the foundational pillar for embodied intelligence, yet a profound disconnect persists between the abstract semantic capabilities of modern Vision-Language Models (VLMs) and the concrete physical constraints required for robust robotic execution. While state-of-the-art models excel in general environmental perception, they frequently generate impractical responses when tasked with operations requiring an understanding of kinematic limits, reachability, or specific motion styles [105]. This deficiency is particularly evident when models fail to enforce user preferences regarding topological constraints or motion trajectories, leading to plans that are semantically correct but physically infeasible [Evaluating VLMs' Spatial Reasoning Over Robot Motion: A Step Towards Robot Planning with Motion Preferences]. The core challenge lies in the lack of inherent understanding of physical laws and robot-specific morphologies within standard VLM architectures, which often treat spatial reasoning as a purely visual or linguistic problem rather than a physically grounded one. Recent critiques suggest that high success rates on standard benchmarks may reflect an "illusion of embodied reasoning," where models exploit lexical-kinematic shortcuts rather than demonstrating genuine cognitive capability regarding physical dynamics [21].

To bridge this gap, recent research has pivoted toward unified representations that abstract robot-specific parameters into generalized spatial forms, thereby enabling generalization across diverse morphologies. A notable approach involves the Space-Physical Reachability Map (S-P Map), which encodes joint limits and kinematic structures into a format independent of specific robot configurations, allowing models like PhysVLM to reason about reachable regions without being tied to a single morphology [105]. This abstraction is critical for generalization, as it enables the model to focus on spatial features rather than learning distinct parameters for every possible robot type. Similarly, other frameworks have sought to bridge the cross-modal semantic gap by abstracting complex visual and spatial information into body-agnostic "Operational Languages," facilitating seamless perception-action coordination across diverse platforms [59]. The concept of "pointing" has further emerged as a unified, embodiment-agnostic intermediate representation, enabling zero-shot generalization across different robots by grounding reasoning in universal visual perception tasks [182]. By treating points, boxes, and traces as optional visual priors, frameworks like GTA-VLA allow human spatial guidance to directly condition the policy's reasoning process, resolving ambiguities that pure language instructions cannot address [10].

The integration of Chain-of-Thought (CoT) reasoning has emerged as a promising strategy to enhance spatial grounding, though its effectiveness varies significantly depending on the implementation and the nature of the physical task. Methods like SpatialCoT utilize coordinate alignment and explicit reasoning steps to improve performance in navigation and manipulation, demonstrating that harnessing the inherent reasoning capabilities of language models can surpass approaches relying solely on point-based action spaces [50]. However, contradictions exist regarding the scalability and utility of explicit CoT in physical domains. Some studies indicate that autoregressive generation of reasoning tokens can introduce latency and compounding errors, suggesting that high-level reasoning alone yields marginal gains unless grounded in concrete linguistic action guidance [8]. Furthermore, while CoT prompting offers benefits in some domains, it provides minimal improvement in tasks heavily reliant on intuitive physical comprehension, such as complex 3D block assembly, indicating that spatial tasks may require more than just verbalized reasoning steps [151]. This suggests a dichotomy where explicit reasoning aids in logical planning but fails to capture the intuitive physics required for fine-grained manipulation.

Despite these advances in representation and reasoning, benchmarks such as ESPIRE and IndustryNav reveal that even frontier models struggle with dynamic environments and active spatial reasoning, often exhibiting high collision rates and failing to plan robust paths in the presence of moving obstacles [116; 288]. Evaluations across diverse benchmarks like Embodied4C and BiManiBench highlight that spatial and temporal reasoning remain primary bottlenecks, with models frequently struggling with dual-arm coordination and mutual kinematic constraints [147; 108]. The reliability of embodied reasoning is often compromised by architectural limitations such as capacity compression and myopic downsampling, which can lead to semantic feature collapse in dynamic scenarios [21]. To mitigate these issues, recent frameworks propose synergistic approaches that integrate high-level reasoning with low-level policy learning, such as Vlasier, which examines how different VLM initializations affect supervised fine-tuning to reduce domain shift [3]. Additionally, the incorporation of physical principles through interaction-based learning, as seen in frameworks like PhysMem, allows agents to refine their understanding of friction and stability at test time, moving beyond static

pre-training data [Learning Physical Principles from Interaction: Self-Evolving Planning via Test-Time Memory]. Comprehensive evaluations suggest that current Multimodal Large Language Models (MLLMs) are not yet fully trustworthy as the central "brain" for robotics, as no single model excels across all essential dimensions of perception, planning, reasoning, and safety [MMRo: Are Multimodal LLMs Eligible as the "Brain" for In-Home Robotics?]. The path forward requires not only better spatial representations but also robust mechanisms for verifying physical feasibility and adapting to the continuous dynamics of real-world environments.

Memory Systems, World Models, and Long-Horizon Task Management

The successful execution of long-horizon tasks in embodied AI necessitates a fundamental architectural shift from reactive, pixel-based prediction to robust memory mechanisms and semantic world models capable of managing complex temporal dependencies. Traditional approaches that rely on reconstructing future frames as pixels often fail to correlate with effective planning decisions, as the objective of visual fidelity can be directly at odds with task relevance [95]. Consequently, recent research posits that world models should instead predict task-relevant semantic information, framing world modeling as a visual question-answering problem that leverages the generalization properties of pretrained vision-language models [95]. This transition is exemplified by the Vision Language World Model (VLWM), which utilizes natural language as an abstract representation of world states to predict trajectories composed of interleaved actions and state changes, enabling both fast reactive planning and reflective reasoning through cost minimization [134]. By compressing raw video into hierarchical structures like Trees of Captions, these models achieve computational efficiency while maintaining the semantic abstraction necessary for high-level task planning, effectively bridging the gap between low-level perception and high-level cognition [134].

However, the reliance on generative rollouts introduces significant challenges regarding physical grounding and reliability. While frameworks like World2VLM successfully distill spatial imagination from generative world models into vision-language models to enable efficient dynamic reasoning, there is a persistent risk that visually plausible simulations may be physically incorrect [World2VLM]. This tension underscores the necessity for "physically viable" world models that represent underlying physical structures to answer intervention queries safely, rather than merely predicting observations that may hallucinate dynamics [Physically Viable World Models]. The complementarity of concrete visual simulation and abstract logical reasoning remains an open area of investigation, with studies suggesting that uncontrolled rollouts can be task-incorrect despite their visual realism, necessitating mechanisms to verify and integrate these simulations with abstract logic [World Models Meet Language Models].

Effective long-horizon management further demands sophisticated memory architectures capable of persisting environmental context and retrieving it for complex reasoning. Standard vision-language models struggle with the vast image collections inherent in multi-day experiences, prompting the development of specialized memory benchmarks and mechanisms that evaluate both recall and action execution [275]. To address spatial complexity, frameworks like 3D-Mem employ multi-view "Memory Snapshots" and frontier-based exploration to construct compact 3D scene representations, allowing agents to reason over both explored and unexplored regions [166]. Similarly, other approaches integrate geometric cues into semantic scene graphs to capture fine-grained spatial-functional relationships, enhancing sequential reasoning in unknown environments [255]. Beyond spatial mapping, memory systems must also support temporal tracking and personalization. Recurrent reasoning frameworks, such as those proposed in recent work on progress estimation, process local video snippets iteratively while maintaining a global Chain of Thought to record task decomposition and completion status, avoiding the computational prohibitions of reprocessing full videos [20]. Furthermore, systems like POLAR and HELPER organize prior interactions into multimodal knowledge graphs or retrieve language-program pairs to resolve user intent and adapt to idiosyncratic procedures over long-term interactions [Personalizing Embodied Multimodal Large Language Model Agents over Long-Horizon User Interactions; 94].

Despite these architectural advancements, significant gaps remain in achieving true autonomy, as evidenced by benchmarks revealing low success rates in household tasks due to difficulties in dependency management and adaptive planning [186]. Evaluations on physical reasoning benchmarks further indicate that current models often fail to translate descriptive knowledge into precise control as task complexity increases [42; 151]. The emergence of "World Action Models" attempts to unify predictive state modeling with action generation, yet the field remains fragmented across learning objectives [226]. Future progress likely depends on hybrid systems that combine the semantic generalization of foundation models with the physical rigor of structured world models, integrating reinforced visual latent planning and adaptive reasoning modes to bridge the gap between high-level cognition and low-level execution [18; 16].

Conclusion

The literature reviewed herein delineates a profound paradigm shift in embodied artificial intelligence, marking a decisive transition from reactive, end-to-end perception-action mappings toward deliberative agents capable of explicit reasoning, long-horizon planning, and self-correction. This evolution is characterized by a move away from monolithic architectures toward sophisticated hybrid frameworks that strategically decouple high-level semantic planning from low-level motor control while maintaining a unified decision-making loop. By integrating mechanisms such as interleaved vision-language reasoning, latent planning, and embodied chain-of-thought, recent advancements have begun to bridge the critical gap between abstract cognitive capabilities and physical execution. These approaches do more than enhance interpretability; they transform agents from passive executors into interactive systems capable of actively resolving ambiguities, verifying their own plans, and recovering from errors in dynamic environments.

Despite these significant strides, the field faces persistent challenges that define the current frontier of research. A central tension remains between the transparency of explicit textual reasoning and the latency requirements of real-time control, prompting a divergence in strategies ranging from verbalizable latent planning to adaptive reasoning triggers. Furthermore, diagnostic evaluations have exposed an "illusion of embodied reasoning," revealing that many state-of-the-art models rely on lexical-kinematic shortcuts rather than genuine causal understanding, often failing under distribution shifts or complex spatial interventions. The disconnect between high-level plans and low-level actions, coupled with the risk of catastrophic forgetting during fine-tuning, underscores the necessity for novel training paradigms that align action representations with pretrained semantic knowledge and enforce physical consistency through reinforcement learning and verifiable rewards.

Looking forward, the trajectory of research points toward the development of holistic Agent AI that unifies perception, reasoning, imagination, and action within shared latent spaces, potentially grounded in bio-inspired cognitive architectures and physically viable world models. The future of embodied intelligence lies not merely in scaling model parameters but in fostering agents that possess intrinsic physical intuition, robust memory systems for long-horizon tasks, and the capacity for social awareness derived from the constraints of physical embodiment. Achieving this vision will require rigorous evaluation protocols that decouple reasoning failures from execution limits, hybrid architectures that balance deliberative slowness with reactive speed, and a fundamental rethinking of how agents learn from and interact with the dynamic, unstructured physical world. Ultimately, true generalization and ethical alignment may emerge not from external reward modeling alone, but from architectures that internalize the vulnerabilities and causal structures of physical existence, enabling agents to operate as trustworthy partners in human environments.